Choose useful axes on an incline

LEARNING TARGET · *R*9 I can use a force diagram to find acceleration on a ramp.

A · KEY VOCABULARY

Parallel ∥	Along the ramp; choose downhill as +.
Normal axis ⊥	Straight out of the ramp, usually not vertical.
Weight parts	$mg \sin\theta$ along; $mg \cos\theta$ into the ramp.
Constraint	No motion through the surface: $a_\perp = 0$.

B · KEY EQUATIONS + UNITS

$$ma_\parallel = mg \sin\theta - f$$

$$N = mg \cos\theta$$

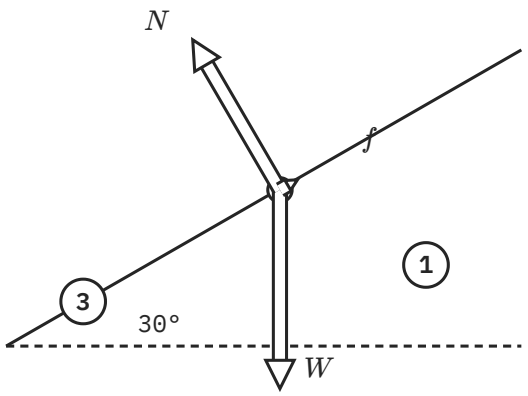
θ ramp angle

N rule no other ⊥ forces

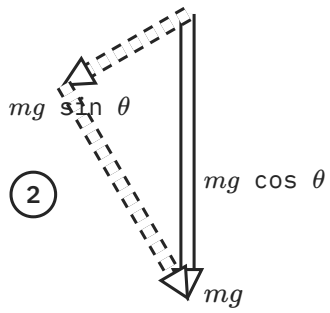
C · COLOR IT · FORCES AND COMPONENTS

Keep the two views separate.

actual forces



resolve W separately



☐ **1 · ORANGE** Actual forces

☐ **2 · ORANGE** Weight's components

☐ **3 · PENCIL** Ramp and angle

D · WORKED EXAMPLE

A **2 kg** block slides down a **30°** ramp with **2 N** friction uphill. **Find its acceleration and normal force.** Use g = **9.8 m/s²**.

GIVEN	EQUATION	WORK	ANSWER
m = **2 kg**; θ = **30°** f = **2 N** uphill $a_\perp = 0$	$mg \sin\theta - f = ma_\parallel$ $N = mg \cos\theta$	$a_\parallel$ = [19.6(0.5) − 2]/2 N = 19.6 cos **30°**	**3.9 m/s²** Downhill; $N \approx$ **17.0 N**.

E · THE 2026-XJ THREAD

DAY 16 OF 22 · NEW TOOL: CHOSEN AXES

A suitable set of axes can simplify the forces changing XJ's path. Components are calculation tools; they must not be counted again as extra forces.

Unit question: How fast is it moving, and where will it arrive?

UNIT 1 COLOR CODE ☐ green speeding up ☐ red slowing down ☐ yellow steady · balanced ☐ blue position · displacement ☐ purple velocity ☐ orange forces pencil · everything else